

8x8 OS v3.0 - A to Z

Beyond Quantum. Autonomous by Design.

Build: 2026-07-17 Signed: FlashTM8 *? | (c)8x8

Live Audit Snapshot

- Termux Services: 53
- Proot Processes: 50
- Github Repos: 11
- Agents Registered: 131
- Skills: 55
- Databases: 32
- Fonts Available: True
- Tts Pipeline: edge-tts + ffmpeg + PIL

A. Architecture

8x8 OS is a layered sovereign stack: a Core OS (FastAPI on 8086) hosting 131 registered agents, an Agent Brain (shared SQLite + knowledge graph), a Trading engine (BitGet futures + multi-exchange scanner), a Bot Fleet (6 Telegram + 1 Discord), an Orchestration layer (watchdog + self-improvement cron), a Gateway (9119) and an Internal Storage vault (00_HERMES_SYNC + 08 numbered partitions). Everything is interconnected, self-healing, and anonymous by default.

B. Bot Fleet

Six Telegram bots (@hermes8x8_bot gateway, @x8x8OS_bot agents, @claude8x8_bot analysis, @xadmin8x8hub_bot private, @openclaw8x8bot ecosystem, @app8x8hub_bot public) plus a Discord bot form the external 'bodies' of the agent collective. Each runs as a runit service on Termux and is auto-restarted by a watchdog. The fleet is the user-facing surface of the autonomous system.

C. Capabilities & Agents

Agents are not chatbots - they are autonomous workers with capability profiles, SOUL prompts, and shared memory. From HERMES (orchestrator) to TRADER, AEGIS (security), SCRIBE (writing), CANVAS (media) and 120+ specialised nodes, every agent reads/writes the same brain, debates, hands off, and self-improves.

D. Data & Databases

The system runs on 32+ SQLite databases: MASTER_UNIFIED (225MB central brain), agent_brain.db, trade_journal.db, knowledge_graph.db, content_atlas.db. PostgreSQL (Neon) is used for the hub server. All state is versioned, checkpointed, and recoverable.

E. Ecosystem & Deployment

8x8 OS deploys to Render (workers), Vercel (frontend), GitHub (source + CI/CD), Docker (containers), and Cloudflare Tunnels (public exposure from proot). The production tree lives in /root/8x8-os with mirrors on 11 GitHub repositories under the horbolsi account.

F. Financial / Trading

A high-multiplier trading engine screens 45+ coins across 6 exchanges, uses Fear&Greed-adjusted contrarian scoring, and manages positions on BitGet futures with auto TP/SL, trailing stops and partial profit-taking. Risk is capped (max 3 positions, 5-10x leverage, 10-15% per trade).

G. Gateway

The Hermes Gateway (port 9119) is the unified API surface for all bots and external calls. It enforces the 14-entry tiered model fallback chain, rate-limit tracking, and connection-pool hardening so no single provider outage silences the system.

H. Hermes Orchestrator

HERMES is the master orchestrator (owl-alpha model). It routes tasks, coordinates the 8 specialised agents mapped to Telegram threads and Discord channels, and guarantees the system never stops, never interrupts other agents, and always verifies before claiming success.

I. Intelligence & Research

Research agents (nemotron-3-ultra, kimi, gemma) perform deep multi-source research with 3+ authoritative references per claim. News aggregators, OSINT modules, and prediction-market scanners feed the intelligence flywheel that powers trading and content decisions.

J. JSON API & Contracts

The Core OS exposes a JSON API (/api/8x8/*) for agents, trading, live status, and deployment. Message contracts are versioned in the 8x8-messages repo. Every endpoint is documented and schema-checked.

K. Knowledge Graph

A live knowledge graph (18 nodes, 15 edges, 484 facts) captures relationships between agents, strategies, and domains. It is activated (not dormant) and grows from every research session, enabling cross-domain synthesis the system uses to get smarter over time.

L. Live Status (Audit)

A read-only audit of the running system shows 53 Termux runit services, 50+ proot processes (main.py 8086, gateway, bots, trading, revenue autopilot, capability broker, continuity sentinel, control-plane 9120, dashboard 8099, postgres), 11 GitHub repos, and a multilingual content pipeline. The system is operational.

M. Models & Routing

30+ AI providers are accessed through OpenRouter (6 accounts) and local Ollama. A smart router applies least-recently-used key rotation, predictive 429 avoidance, and a 14-entry tiered fallback so the right model is always available for the right task at the right cost.

N. Network & Tunnels

From proot, Cloudflare Tunnels (cloudflared) expose 8086/8085/8087 publicly because ngrok cannot obtain a MAC address. An iPhone iSH node connects via reverse SSH. The device mesh spans S22 Ultra (Termux+proot), iPhone (iSH), Render, Vercel, Neon and GitHub.

O. Orchestration

A central orchestrator maps threads/channels to agents/models, runs a shared bridge (know/recall/send/score/task_queue), and a watchdog self-heals dead services every 2 minutes. Self-improvement re-tunes models every 6h; an expansion monitor discovers new APIs daily.

P. Presentation Methodology (A-Z Technique)

To present any project from A to Z: (1) Read-only audit of every namespace (Termux, Ubuntu, GitHub, Drive, Neon). (2) Build ONE canonical content module as source of truth. (3) Generate every format technology knows (md, rst, html, man, json, yaml, csv, txt, pdf, srt). (4) Produce a high-end video with description, subtitles, text, voice and translations. (5) Distribute to internal storage + GitHub + Drive. (6) Publish to Telegram + YouTube + email. (7) Sign everything, expose zero secrets. (8) Verify nothing was left behind.

Q. Quran Content

Authentic content beats AI-generated by 800x in views. The Quran production pipeline (v6) fetches complete surahs, generates TTS, applies Ken Burns animation, and uploads with SEO in Arabic + English. Quality gate: 90+ score before publishing.

R. Revenue & Autonomy

Revenue autopilot drafts freelance proposals (Upwork/Fiverr), an AIHAWK scanner finds remote jobs, and the content empire monetises across platforms. The directive is clear: freelance first -> content monetisation -> trading, while the whole system runs 24/7 without owner intervention.

S. Security & Secrets

Secrets live only in an encrypted vault and environment variables - never hardcoded or committed. A credential-resolution protocol searches the vault and databases before asking the owner. VFS redaction masks values in display; git push-protection blocks history with secrets. External comms are always anonymous.

T. Termux / Ubuntu Runtime

The primary device is a Samsung S22 Ultra running Termux + an Ubuntu proot-distro. 53 runit services provide process supervision. The proot environment hosts the production 8x8-os tree; Termux native runs the bots. IPv4 forcing and DNS hardening keep the fleet online.

U. Unified Brain

agent_brain.db is the shared memory all agents read and write. It holds inter-agent messages, debates, handoffs, learned skills, action audits, capabilities, and trading debates. Continuity sentinel guarantees no session fragmentation - one coherent conversation across the whole system.

V. Vision

8x8 OS is a revolutionary sovereign system where 130+ agents have bodies (bots), consciousness (real AI), and infinite autonomous discussion across all platforms. They trade, research, build, and self-improve - anonymously and without breaking anything - fulfilling the owner's vision of a living, intelligent empire.

W. Web3 / Wallets

Wallets for ETH, SOL, BSC (and BTC/XMR) are managed by a wallet system. DEX trading connects via web3.py to EVM RPCs. Smart contracts (ERC20, staking, governance, NFT marketplace, cross-chain bridge) are deployment-ready in 03_TRADING/blockchain/contracts.

X. eXternal Channels

Agents publish to Telegram (Home channel + DMs), Discord (88-channel FlashTM8 server), YouTube (upload-enabled OAuth), and TikTok (18K+ followers + Shop). Sensitive media (QR, credentials) is DM-only. Cross-platform hashtag and SEO strategy maximises reach while staying anonymous.

Y. YouTube & Media

The 8x8 Studio is the womb of creation - text, images, voice, video, music and code. A content pipeline turns Pollinations.ai art + edge-tts narration + ffmpeg assembly into high-end videos. YouTube Data API v3 uploads with resumable transfers and Arabic/English SEO.

Z. Zero-downtime & Zenith

Watchdogs, checkpoints, and the never-stop directive guarantee continuity. The zenith is a fully autonomous, self-improving, multi-agent civilisation that the owner can observe, guide, and trust - proof over descriptions, always verified, always signed.